

Project Design Phase – Solution Architecture

Project: Book a Doctor App

Date: 18 June 2026

Architecture Overview

Frontend: React + Bootstrap UI with routes for patient, doctor, and admin modules.

Backend: Node.js + Express REST APIs for authentication, doctors, appointments, documents, and admin operations.

Database: MongoDB collections for Users, Doctors, Appointments, and uploaded file metadata.

Security: JWT auth, bcrypt hashing, role-based middleware, request validation, and protected routes.

Core Flow

Patient/Doctor/Admin logs in → JWT issued → frontend stores auth state → requests protected APIs → backend validates token/role → reads/writes MongoDB → returns JSON response → UI updates dashboards and notifications.